

SONGKHLA, Thailand

WMO: 485680

Lat: 7.187N

Lon: 100.608E

Elev: 12

StdP: 14.69

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	74.3	75.0	67.7	101.9	81.5	68.9	106.3	81.9	16.8	81.8	15.3	82.0	2.2	240	0.355

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	11.6	93.5	78.6	92.4	78.8	91.4	78.9	81.6	89.2	80.9	88.6	80.4	88.1	7.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.2	151.7	86.5	78.9	149.8	86.2	78.3	146.9	85.5	45.3	89.6	44.7	89.0	44.1	88.7	89.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.2	12.4	11.4	DB	71.7	97.3	1.9	2.5	70.3	99.1	69.2	100.6	68.2	102.0	66.8	103.8	
			WB	68.8	84.0	2.1	1.9	67.2	85.3	66.0	86.5	64.8	87.5	63.3	88.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	82.9	81.1	81.8	82.9	84.3	84.9	84.5	84.1	84.0	83.4	82.2	81.0	80.8
	DBStd	2.26	1.73	1.60	1.66	1.60	1.63	1.76	1.78	1.94	1.63	1.76	1.94	1.98
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12018	965	889	1019	1029	1081	1036	1058	1054	1003	999	931	954
	CDD65	6543	500	469	554	579	616	586	593	589	553	534	481	489
	CDH74	73548	5155	5097	6478	7257	7555	6930	6904	6722	6095	5601	4852	4902
	CDH80	26195	1209	1420	2327	3115	3276	2884	2802	2702	2250	1800	1266	1144
Wind	WSAvg	4.3	6.0	6.1	5.0	4.1	3.5	3.5	3.7	3.9	3.7	3.5	3.8	5.2
Precipitation	PrecAvg	85.2	4.5	1.2	2.0	3.1	4.6	4.1	3.6	4.7	5.0	11.3	21.6	17.4
	PrecMax	132.1	26.9	4.9	20.7	10.7	11.8	12.4	7.4	14.4	12.8	22.8	52.5	47.8
	PrecMin	52.3	0.1	0.0	0.0	0.0	0.0	0.4	0.0	0.2	0.6	3.1	1.8	2.3
	PrecStd	18.8	5.0	1.3	3.1	2.5	2.2	2.4	1.9	3.1	2.6	4.4	9.6	10.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	88.1	89.4	91.4	93.2	94.5	94.5	94.4	94.9	93.2	91.7	89.5	88.2
		MCWB	78.2	77.7	78.9	80.6	79.4	78.5	77.9	77.7	78.3	78.3	79.0	78.7
	2%	DB	86.7	87.7	89.6	91.4	92.8	92.7	92.6	93.0	91.5	89.8	87.7	86.6
		MCWB	77.9	78.1	78.8	80.2	79.9	78.8	78.2	77.8	78.4	78.3	79.0	78.5
	5%	DB	85.7	86.3	88.3	90.1	91.3	91.2	91.2	91.3	89.9	88.1	86.4	85.8
		MCWB	77.4	77.5	78.6	79.7	79.7	79.0	78.3	78.1	78.5	78.6	78.6	78.2
	10%	DB	84.5	85.7	87.4	89.3	90.0	89.7	89.7	89.6	88.4	87.0	85.5	84.6
		MCWB	76.8	77.2	78.4	79.5	79.6	79.0	78.3	78.1	78.3	78.4	78.3	77.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.0	79.9	80.9	82.3	82.5	81.9	81.0	81.0	81.0	80.8	80.8	80.7
		MCDB	86.0	86.6	88.2	90.4	90.6	89.6	89.1	89.2	88.9	87.4	86.6	85.6
	2%	WB	79.0	79.0	79.9	81.3	81.5	80.9	80.1	80.0	80.0	79.9	79.8	79.5
		MCDB	84.9	85.9	87.4	89.2	89.6	88.9	88.3	88.4	88.0	86.7	85.7	84.8
	5%	WB	78.3	78.3	79.3	80.7	80.8	80.1	79.5	79.4	79.4	79.3	79.2	78.9
		MCDB	84.0	85.1	86.7	88.4	88.9	88.2	87.9	87.9	87.3	86.0	84.8	84.2
	10%	WB	77.6	77.7	78.9	80.0	80.2	79.5	78.9	78.8	78.8	78.8	78.7	78.2
		MCDB	83.3	84.4	86.2	87.6	88.0	87.5	87.4	87.3	86.7	85.5	84.2	83.3
Mean Daily Temperature Range	5% DB	MDBR	7.5	8.5	9.7	10.6	11.6	12.1	12.3	12.6	11.7	10.5	8.3	7.4
		MCDBR	8.5	9.4	10.9	11.7	13.0	13.5	14.1	14.4	13.5	12.1	10.0	8.9
		MCWBR	4.1	4.2	4.3	4.6	4.9	4.8	4.7	4.7	4.7	4.8	4.3	4.4
	5% WB	MCDBR	8.0	9.0	10.2	11.2	12.0	12.3	12.6	13.0	12.2	11.1	9.3	8.5
		MCWBR	4.2	4.4	4.4	4.8	5.2	5.2	5.3	5.3	5.1	4.9	4.4	4.5
Clear-Sky Solar Irradiance	taub	0.467	0.468	0.476	0.461	0.447	0.454	0.460	0.456	0.447	0.447	0.458	0.470	
	taud	2.281	2.258	2.261	2.312	2.375	2.383	2.358	2.351	2.373	2.379	2.345	2.291	
	Ebn at Noon	270	273	271	271	269	263	263	268	275	275	271	266	
	Edh at Noon	43	45	45	43	39	38	39	40	40	39	40	42	
All-Sky Solar Radiation	RadAvg	1512	1822	1866	1847	1659	1587	1571	1616	1582	1463	1293	1268	
	RadStd	116	101	140	87	87	70	73	55	58	101	121	149	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	+0.36	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+216	+218
(47) Regional (15 neighbors)	+0.34	N/A	N/A	+0.56	+0.31	+0.34	N/A	N/A	+158	+160

Nomenclature: See separate page